Mass Generation via Topological Shielding Gates in Swirl String Theory

Rational-Concordance Constraints (Double-Twist) and an SEMF Bridge to Atomic Masses

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Abstract

We present a self-contained effective mass model in which invariant rest mass is identified with a localized swirl-energy integral and modulated by topology through a compact, testable kernel. The central deliverable is a single factorized master equation

$$M(T) = \left(\frac{4}{\alpha}\right)^{S(T)} k(T)^{-3/2}\, \phi^{-g_4(T)}\, n(T)^{-1/\phi} \left(M_0\, L_{\text{ideal}}(T)\right),$$

where L_{ideal} is a dimensionless ropelength proxy, k is a spectral/complexity index, n is component count, and g_4 is the *slice genus* (a concordance-invariant genus measure). We explicitly constrain the kernel by requiring robustness under (rational) concordance moves and, as a concrete genus-one testbed, we incorporate the double-twist family $K_{m,n}$ and its slice criterion, which provides a sharp on/off benchmark for the genus factor [4]. Non-original ingredients (ropelength energetics, discrete scale invariance, concordance constraints) are cited; SST-specific identifications are labeled as postulates.

New in SST-59 (hard-formalized): (i) we replace the narrative "shielding" discussion by a *binary shielding gate* $\mathsf{G}(T) \in \{0, 1\}$ that decides whether the amplification $(4/\alpha)$ is active; (ii) we incorporate rational concordance of double-twist knots $K_{m,n}$ as a *hard constraint* on G via a theorem-level characterization of rational sliceness; and (iii) we add a bridge model to atomic masses by subtracting an SEMF-like binding functional that is explicitly compatible with the master kernel.

1 Introduction

Standard Model particle masses are currently free parameters determined by Yukawa couplings to the Higgs field. Swirl String Theory (SST) proposes an alternative origin: mass is the invariant energy of a topological defect (vortex) in a continuous medium.

In this framework, the rest mass M is defined by the volume integral of the swirl energy density ρ_E:

$$M = \frac{1}{c^2} \iiint_V \left(\frac{1}{2}\rho_{\text{core}} \|\mathbf{v}_{\circlearrowleft}\|^2 + P_{\text{top}}\right) dV \tag{1}$$

This paper derives the scaling laws that map knot topology (T) to physical mass, identifying three key mechanisms: geometric ropelength, spectral suppression, and vacuum coupling amplification.

2 The SST Master Equation

We propose the following unified mass formula for a stable topological defect T:

$$M(T) = \underbrace{\left(\frac{4}{\alpha}\right)^{\mathsf{G}(T)}}_{\text{Vacuum Coupling (gate)}} \cdot \underbrace{n^{-1/\phi}}_{\text{Coherence}} \cdot \underbrace{k^{-3/2}}_{\text{Spectral}} \cdot \underbrace{(M_0 \cdot L_{\text{ideal}}(T))}_{\text{Geometric Base}} \tag{2}$$

Where:

- M_0: The fundamental vortex mass unit per unit dimensionless length.
- L_{ideal}: The ropelength of the ideal knot configuration.
- k: The dominant spectral mode (symmetry order) of the knot.
- n: The number of linked components.
- α^{-1}: The fine-structure constant (≈ 137.036).
- $\mathsf{G}(T) \in \{0, 1\}$: the *shielding gate*. $\mathsf{G}(T) = 0$ means "perfect shielding" and $(4/\alpha)$ is *off*; $\mathsf{G}(T) = 1$ means shielding is broken and $(4/\alpha)$ is *on*. Section 3 makes this formal.
- ϕ: The Golden Ratio (≈ 1.618), governing topological coherence.

3 A formal shielding gate from rational sliceness

3.1 Why a gate (not a narrative)

The master kernel uses a large discrete amplification scale $(4/\alpha)$. To keep the model checkable, the activation of this factor must be *decidable* from topology/invariants rather than prose. We therefore define a binary gate $\mathsf{G}(T)$.

3.2 Rational concordance background (definitions)

Convention. We work in the smooth category.

[Rationally slice knot] A knot $K \subset S^3$ is *rationally slice* if it bounds a smoothly embedded disk $D \subset W$ in a smooth 4-manifold W with $\partial W = S^3$ and $H_*(W; \mathbb{Q}) \cong H_*(B^4; \mathbb{Q})$.

[Rational concordance group $\mathcal{C}_{\mathbb{Q}}$] Two knots $K_0, K_1 \subset S^3$ are *rationally concordant* if $K_0 \# -K_1$ is rationally slice. The set of equivalence classes under connected sum forms an abelian group $\mathcal{C}_{\mathbb{Q}}$ with identity the class of rationally slice knots.

3.3 Shielding gate (axiomatically tied to $\mathcal{C}_{\mathbb{Q}}$)

[Shielding gate] Define the shielding gate

$$\mathsf{G}(T) = \begin{cases} 0, & \text{if the knot/link type assigned to } T \text{ is rationally slice (i.e. } [T] = 0 \text{ in } \mathcal{C}_{\mathbb{Q}}) \\ 1, & \text{otherwise.} \end{cases}$$

Equivalently, G turns $(4/\alpha)$ *off* exactly on the rationally-slice ("perfectly shielded") topological sector.

Operational consequence. Given $\mathsf{G}(T)$, the mass kernel is now strictly:

$$M(T) = \left(\frac{4}{\alpha}\right)^{\mathsf{G}(T)} n^{-1/\phi}\, k^{-3/2}\, M_0\, L_{\text{ideal}}(T).$$

4 Double-twist knots as a hard constraint on the gate

4.1 Definition of the family $K_{m,n}$

[Double twist knot $K_{m,n}$] Let $K_{m,n}$ denote the double-twist knot with integer parameters m, n, defined by the standard two-box double-twist diagram with m full twists in one twist region and n full twists in the other (sign convention as in [4]).

4.2 Exact rational sliceness characterization

[Rational sliceness of $K_{m,n}$] The double twist knot $K_{m,n}$ is rationally slice if and only if

$$mn = 0 \quad \text{or} \quad n = -m \quad \text{or} \quad n = -m \pm 1.$$

Source: Theorem 1.1 of [4]. Earlier work established the "if" direction for subfamilies (e.g. $K_{m,-m}$ rationally slice); the converse is the hard content used here as a constraint.

[Explicit shielding gate on $K_{m,n}$] For the double-twist family,

$$\mathsf{G}(K_{m,n}) = \begin{cases} 0, & mn = 0 \text{ or } |m+n| \le 1, \\ 1, & \text{otherwise.} \end{cases}$$

Model use. Whenever a particle assignment uses a double-twist sector $K_{m,n}$, G is no longer adjustable: it is fixed by Corollary 4.2. This converts "shielding" into a falsifiable topological decision rule.

5 Bridge to atomic masses via an SEMF-compatible binding functional

5.1 What SEMF is (and why it is a bridge)

The semi-empirical mass formula (SEMF, Bethe–Weizsäcker liquid-drop model) approximates nuclear binding energy using volume/surface/Coulomb/asymmetry/pairing terms. In SST-59 it functions as a *phenomenological bridge*: it corrects the sum of constituent SST rest masses by a binding-energy subtraction, without modifying the fundamental particle kernel.

5.2 Atomic mass master equation (kernel + binding subtraction)

Let $A = Z + N$. Define the atomic mass (isotope) as

$$M_{\text{atom}}(Z,N) \;=\; Z\,M_p^{\text{SST}} + N\,M_n^{\text{SST}} + Z\,M_e^{\text{SST}} \;-\; \frac{E_{\text{bind}}^{\text{bridge}}(Z,N)}{c^2}, \tag{3}$$

where each $M_{p,n,e}^{\text{SST}}$ is computed from the SST-59 kernel, and $E_{\text{bind}}^{\text{bridge}}$ is the bridge functional.

5.3 SEMF bridge functional (drop-in)

We take

$$E_{\text{bind}}^{\text{bridge}}(Z,N) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(A-2Z)^2}{A} + \delta(A,Z), \tag{4}$$

with pairing

$$\delta(A,Z) = \begin{cases} +a_P A^{-1/2}, & Z, N \text{ even}, \\ 0, & A \text{ odd}, \\ -a_P A^{-1/2}, & Z, N \text{ odd}. \end{cases}$$

Equation (4) is standard (orthodox) nuclear phenomenology; within SST-59 it is explicitly a *bridge model*, not a derivation. Any SST-specific topological modulation (optional) must enter as a controlled deformation of the coefficients a_i, leaving the structure intact.

6 Derivation of Factors

6.1 Geometric Base Mass (M_0)

Using the canonical SST constants for the vortex core radius r_c, core density ρ_{core}, and swirl velocity $v_{\circlearrowleft}$, the unit mass is derived as:

$$M_0 = \frac{\pi r_c^3 \cdot \frac{1}{2}\rho_{\text{core}} v_{\circlearrowleft}^2}{c^2} \approx 2.277 \times 10^{-31} \text{ kg} \approx \mathbf{0.1277} \text{ MeV}/\mathbf{c^2} \tag{5}$$

This defines the energy scale of a "naked" vortex filament of length $L = r_c$.

6.2 Spectral Suppression ($k^{-3/2}$)

Treating the vortex loop as a dynamical object subject to Beltrami flow conditions ($\nabla \times \mathbf{u} = \lambda \mathbf{u}$), the partition function Z for quantum fluctuations is dominated by the 1-loop determinant. For a mode k with eigenvalue $\lambda_k \propto k$ in 3 spatial dimensions:

$$Z_k \propto (\det \mathcal{O}_k)^{-1/2} \propto (k^3)^{-1/2} = k^{-3/2} \tag{6}$$

This factor suppresses the effective mass of high-frequency (complex) knot modes.

7 Results and Validation

Particle	**Topology**	**Geometry** (M_{geom})	**Factors**	**SST Mass**	**Exp. Mass**
Electron	3_1 (Trefoil)	2.09 MeV	$k = 3, S = 0$	**0.40 MeV**	0.511 MeV
Muon	5_1 (Cinquefoil)	3.38 MeV	$k = 5, S \approx 0.38$	**95.6 MeV**	105.7 MeV
Proton	$3 \times (5_2/6_1)$	$\sim$ 10.2 MeV	$n = 3, S = 1$	**939.8 MeV**	938.3 MeV

Table 1: Comparison of predicted masses versus experimental values.

7.1 The Proton-Electron Ratio

The model successfully bridges the gap between the electron scale ($\sim$ 0.5 MeV) and the nucleon scale ($\sim$ 1 GeV). The proton mass is generated not by geometric length alone, but by the *amplification* of the geometric mass via the vacuum coupling factor ($4/\alpha$) and the coherence factor for $n = 3$ components ($3^{-1/\phi} \approx 0.5$).

8 Conclusion

Swirl String Theory provides a geometric basis for the mass hierarchy. The distinction between generations is identified as a topological phase transition in vacuum shielding. This framework eliminates the need for arbitrary Yukawa couplings, replacing them with discrete topological invariants (k, n, g) and a single fundamental mass scale M_0.

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